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# Sajjil Akhtar

[email placeholder] | [location placeholder] | [LinkedIn placeholder] | [GitHub: merklenode]  
(<https://github.com/merklenode>)

## Summary

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Passionate AI&ML student and accomplished Blockchain Developer with a strong foundation in full-stack development. Proven ability to design, build, and deploy robust applications, including a production-ready e-commerce platform with Supabase integration. Eager to leverage expertise in emerging technologies to solve complex problems and contribute to innovative projects.

## Skills

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**Languages:** TypeScript, JavaScript, Python, C/C++ (inferred) **Web Technologies:** Node.js, React, Next.js, RESTful APIs, UI/UX Design **Databases:** Supabase (PostgreSQL) **Cloud/Deployment:** Vercel, Git, GitHub Actions, CI/CD **Blockchain:** Blockchain Architecture, Merkle Trees, Proof-of-Work (PoW), Smart Contracts, Cryptocurrencies **AI/ML:** Machine Learning Concepts, AI Protocol Design, Agent-based Systems **Methodologies & Tools:** Full-Stack Development, System Design, Code Architecture, Project Management, Agile Principles, Refactoring

## Projects

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**FreshTerra (Full-Stack E-commerce Platform)** \* Designed, developed, and deployed a multi-phase e-commerce storefront with a comprehensive administrative dashboard. \* Achieved production readiness for a Supabase launch, including architecting and integrating the backend for Supabase-only storage. \* Implemented key features such as product management (add, edit, delete), order fallback, and an intuitive admin UI. \* **Tech Stack:** TypeScript, Next.js (inferred), React (inferred), Supabase, Node.js (inferred), Vercel

**merkleNode (Merkle Tree Implementation & Portfolio Site)** \* Developed a robust Merkle Tree implementation and restructured a portfolio site with a focus on UI primitives and navigation configuration. \* Integrated CI workflows using GitHub Actions to ensure code

quality and deployment efficiency. \* **Tech Stack:** TypeScript, Next.js (inferred), React (inferred), GitHub Actions, Vercel

**Agent-s (Model Context Protocol Prototype)** \* Prototyped an open and extensible Model Context Protocol (MCP) designed to connect AI-powered applications with diverse data sources, APIs, and blockchain-based smart contracts. \* **Tech Stack:** TypeScript, AI/ML Concepts, Blockchain Concepts

**Blockchain Prototypes (Proof-of-Work & Cryptocurrency Systems)** \* Implemented various blockchain prototypes, including a Simple Blockchain Prototype, Cryptocurrency System, and Multithreaded Block Mining simulations. \* Explored fundamental blockchain concepts such as Proof-of-Work consensus mechanisms and distributed ledger technology. \* **Tech Stack:** C/C++ (inferred), Makefile, Blockchain, Cryptography, Multithreading

## Experience

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### Software Developer (Self-Directed Projects) [Current Month, Year] – Present

\* Led the full-stack development of FreshTerra, an e-commerce platform, from initial storefront to a comprehensive administrative dashboard, achieving production readiness for Supabase deployment. \* Architected and integrated a Supabase backend, establishing a robust and scalable data solution for e-commerce operations. \* Designed and implemented key administrative functionalities, including efficient product management (add, edit, delete), order processing, and an intuitive user interface for the admin panel. \* Developed and refined Merkle Tree implementations and explored advanced blockchain concepts, including Proof-of-Work (PoW) consensus and multithreaded block mining. \* Prototyped an innovative Model Context Protocol (MCP) to bridge AI applications with various data sources, APIs, and blockchain smart contracts. \* Applied best practices in code architecture, refactoring, and CI/CD (GitHub Actions) to ensure system scalability, maintainability, and efficient deployments.

## Education

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